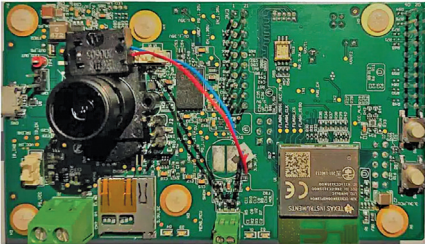


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4K and Fast AF Camera	Low-Power Wireless Camera	Ultra-Compact Camera Cube
		
The reference design for a 4K, fast autofocus camera with AI functionality, featuring high dynamic range (HDR) and PDAF assist, suitable for security applications. The RAA462113FYL is a reference design by Renesas compatible with MIPI-CSI2 and SLVS output formats. It can support up to 60fps with 12-bit digital output, making it a versatile choice for advanced security imaging needs. The design significantly reduces the development time for 4K cameras with fast autofocus and AI capabilities, allowing developers to expedite hardware development using the provided PCB design and streamline autofocus software creation with available AF sample software. Additionally, leveraging AI applications from RZ/V partners shortens the AI model development period, accelerating the creation and implementation of advanced camera systems. This technology stands out with its 4K video output, enabled by the RAA462113FYL, a 4K CMOS Image Sensor with PDAF pixels, and a Novatek image signal processor for fast autofocus using a Hybrid system of PDAF and Contrast AF. The high-speed AI inference processing is powered by Renesas' DRP-AI on the RZ/V platform, offering high efficiency in the 1TOPS/W class, thereby enhancing performance and energy efficiency in AI tasks. The reference design is suitable for use in network surveillance cameras and network video conference cameras. Renesas https://www.electronicsforu.com/electronics-projects/reference-design-for-4k-and-fast-af-camera	The reference design for a low-power wireless camera presents a complete solution for a battery-operated wireless camera, combining components in image processing, connectivity, sensor technology, and power management. The TIDA-010224 is a reference design by Texas Instruments (TI). It utilises high-efficiency power converters and an optimised system architecture to extend battery life significantly. The design incorporates an advanced image processor and a Wi-Fi connectivity module, facilitating secure, high-quality live video streaming and two-way audio streaming, with features like noise reduction and echo cancellation. Additionally, the design is equipped with infrared (IR) LEDs and IR cut filters, enabling the camera to function effectively in low-light conditions, thus offering robust night vision capabilities. The design significantly reduces new product development cycles by integrating features like ambient light sensing, an IR LED driver, and a motorised IR cut filter for optimal day and night functionality. It includes SimpleLink Wi-Fi for easy control over various networks and supports multiple frequencies with its concurrent multiprotocol feature. The power architecture is versatile, accommodating both USB and battery sources. Focused on low power consumption, it incorporates efficient DC/DC converters to extend battery life and offers various wake-up options for added flexibility. Wi-Fi components are optimised to reduce power usage significantly, and sub-1GHz technology integrates with sensors for automatic video recording, making the design both power-efficient and highly functional. The reference design is particularly well-suited for use as a wireless security camera and video doorbell. Texas Instruments (TI) https://www.electronicsforu.com/electronics-projects/low-power-wireless-camera-reference-design	The camera cube reference design showcases the feasibility of AI functioning on a low-power budget, allowing time-sensitive and safety-critical applications to run on small batteries. The MAXREFDES178 is the reference design by Analog Devices that includes the MAX78000, a low-power microcontroller with a neural network accelerator for audio and video inferences. It also features the MAX32666, an ultra-low power Bluetooth microcontroller, and two MAX9867 audio CODECs. This system is provided in a compact form to demonstrate the embedding of AI applications like facial identification and keyword recognition in low-power, cost-effective settings like wearables and IoT devices. The AI accelerator in the MAX78000 is designed to reduce the power required for AI inferences in vision and hearing applications by up to 1000 times compared to other embedded solutions, according to the company. Additionally, AI inferences on the MAXREFDES178# demonstrate significant latency improvements, performing over 100 times faster than on a standard embedded microcontroller. The camera cube's compact size, measuring 4.1cm x 4.3cm x 3.8cm, illustrates the feasibility of integrating AI into wearables and space-limited IoT devices. Maxim highlights that the MAX78000 solution is about 50 percent smaller than the smallest comparable GPU-based processor and eliminates the need for additional components like external memories or complex power supplies, enabling more cost-effective AI inferences. The reference design is suitable for wearables and IoT devices. Analog Devices (ADI) https://www.electronicsforu.com/electronics-projects/camera-cube-reference-design